

TRYPTOPHAN TOXICITY—TIME AND DOSE RESPONSE IN RATS

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ABSTRACT

During the past decade L-tryptophan (Trp) ingestion have been associated with a multi-systemic syndrome, known as eosinophilia myalgia syndrome (EMS). Even though an epidemic studies indicated that a contaminant 1,1'-ethyldene-bis-L-tryptophan was involved in EMS, abnormalities in metabolism of Trp have been reported in other similar clinical syndromes such as carcinoid syndrome, scleroderma or eosinophilic fasciitis.

The purpose of the study was to investigate the role of Trp or its metabolite, given in different dosing regimens in induction of tissue damage.

Method: 3 months old female rats (Charles River CD-1) were fed for 3,6,12 weeks on a diet containing 20% protein diet derived from casein and supplemented with 1%, 2%, or 5% Trp. On the last week of feeding, half of the animals fed on a control diet and half of the animals fed on the Trp diet were injected with 2 injections of para-chlorophenyl alanine (p-CPA), a Trp hydroxylase inhibitor, 300 mg/kg i.p followed by 3 injection of 100 mg/kg every alternate day.

Results: Body weight of rats fed higher levels of Trp increased slowly and injection of p-CPA induced loss in body weight. 2/6 of the animals treated with 1% Trp and 1/6 treated with 5% Trp for 3 weeks and 2/4 animals treated with 1% Trp and 1/4 treated with 5% Trp for 12 weeks died after injection of p-CPA. No mortality was detected in 1–5% Trp treated animals. Alopecia and skin changes were seen after p-CPA in 1–5% Trp treated animals. Increased amounts of connective tissue and induction of inflammatory cell proliferation were observed in lung, spleen and in gastrocnemius muscle of rats treated with higher dose of Trp for longer period. Induction of kynurenine pathway by injection of p-CPA caused more tissue damage.

It is concluded that excessive Trp or elevation of its metabolites could play a role in amplifying some of pathological features of EMS. This pathological damage is further augmented by metabolites of the kynurenine pathway.

1. INTRODUCTION

Tryptophan (Trp) is used as a food additive due to its being an essential aromatic amino acid and a precursor of niacinic acid. It is essential for most proteins biosynthesis, is a unique source for serotonin (5HT), an important neurotransmitter in the central nervous system and an immune modulator and vasoactive amine in periphery. Additionally it is also a major source of the niacin amide coenzyme NAD. Excess Trp in the diet is normally degraded by liver enzyme Trp 2,3 dioxygenase (TDO) or by extrahepatic inducible enzyme, indoleamine 2,3 dioxygenase (IDO) which further metabolizes it in the kynurenine pathway producing nicotinic acid, xanthurenic acid and picolinic acid (Cooper *et al.*, 1991). Trp has been widely marketed as a non-prescription drug for insomnia depression or anxiety in large doses up to 15 grams. During the past decade it has been reported that diets containing excessive L-Trp have caused a multi-systemic syndrome, known as eosinophilia myalgia syndrome (EMS) (Duffy, 1992). Even though an epidemic study indicated that a contaminant 1,1 ethylidenebis L-tryptophan (EBT) was involved in EMS (Mayeno *et al.*, 1990), it is not unlikely that the contaminate per se would be able to elicit this syndrome. Abnormalities in the metabolism of tryptophan were also reported to be associated with tissue fibrosis and inflammation in syndromes, such as carcinoid syndrome and scleroderma-like syndrome (Freis *et al.*, 1973), eosinophilic fasciitis (Bruce *et al.*, 1990; Varga *et al.*, 1990) and after therapy with Trp and carbidopa (Sternberg *et al.*, 1980). Serotonin, a metabolite of tryptophan, has a growth stimulatory effect on several cell types and has a mitogenic effect on endothelial cells (Pakala *et al.*, 1994). The immune system is also a target for many toxicants, including those obtained through products of Trp degradation such as kynurenine, quinolinic acid or picolinic acid (Leonhardt *et al.*, 1986; Freese *et al.*, 1990; Heyes *et al.*, 1992b; Varesio *et al.*, 1990; Saito *et al.*, 1993). Animal studies results have been controversial. It has been reported previously that excessive dietary Trp and induction of kynurenine pathway resulted in appearance of increased fibrosis and increase of TNF- α positive cells (Ronen *et al.*, 1996), other groups have reported that only Trp implicated in EMS epidemic provoked myofascial fibrosis in rats (Silver *et al.*, 1994; Love *et al.*, 1993). In view of the variable action of Trp it was the purpose of the study to investigate if L-Trp or its metabolites, given in different dosing regimens may play a role in the development of a toxic effect.

2. MATERIALS AND METHODS

Three-month-old female rats (Charles River CD-1) were divided into groups of 6 animals each, fed at various time interval from 2 days to 12 weeks with 20% protein diet derived from casein, with or with no supplementation of 1%, 2% and 5% Trp (Sigma, St Louis, MO, USA). Animal fed 1% Trp were sacrificed at 2 days, 3, 6 and 12 weeks. Animals fed 2% Trp were sacrificed at 2 days, 3 and 6 weeks and animals fed 5% Trp were sacrificed after 3 and 12 weeks. On the last feeding week, half of the animals fed on the control diet and half of the animals fed on the Trp diet were injected with para chlorophenyl alanine

(p-CPA), an inhibitor of Trp hydroxylase. Two injections of 300 mg/kg 24 h apart, followed by three additional injections of 100 mg/kg on every alternate day were given intra peritoneal (i.p.). Animals were kept in conditions of controlled temperature (24°C), maintained on a 12–12 h light-dark cycle. Food and water were supplied *ad libitum*. Food intake was calculated for each animal on every alternate day and body weight was measured weekly. At the end of the feeding period, animals were sacrificed and tissue samples of gastrocnemius muscle, spleen, lung and liver were collected. The samples were fixed in 10% natural phosphate buffered formaldehyde, embedded in with May-Grunwald Giemsa for detection of mast cells and eosinophiles, or Mallory stain for fibrosis. Frozen section of muscle and lung were collected for immunohistochemical demonstration of tumor necrosis factor- α (TNF- α) using polyclonal rabbit anti rat TNF- α and streptavidin—biotin peroxidase conjugate (Zymed laboratories, South San Francisco, CA). In addition, two animals from each group were injected i.p. three hours prior to sacrifice, with 2 (Ci/gr body weight of 3H-thymidine, (50 Ci/mol specific activity, Nuclear Research Center, Beer Sheva, Israel). Tissue of lung and muscle were collected for autoradiography.

For all the measurement statistical analysis was performed using unpaired student's t-test and results were reported at mean $\pm$ S.E.M. Analysis of variance (ANOVA) for all groups was performed and $p < 0.05$ was considered as significant value.

3. RESULTS

Clinical observations included significant reduction in growth pattern in animals fed excessive L-Trp after 12 weeks of feeding. Injection of p-CPA induced significant loss in body weight compared to control ($p < 0.01$) (Fig. 1). Food consumption was not different in all groups. Mortality was significantly higher in rats fed Trp and injection of p-CPA ($P < 0.05$). 1/3 of rats fed 1% Trp and 1/5 of animals fed 5% Trp died. No mortality was detected in control, Trp or p-CPA-treated groups. In addition, over 50% of rats fed with 2% and 5% Trp developed alopecia and skin changes after injection of p-CPA. Table 1 summarizes the morphological changes observed in gastrocnemius muscle of all experimental groups. Administration of tryptophan or injection of p-CPA alone were accompanied by mild inflammatory infiltration composed from macrophage, monocytes and frequent mast cells. Moderate hyperplasia of fibroblasts was noted in endomysial septa and in some animals mild endothelial cell hyperplasia was present. It seemed to be more severe and more frequent in animals exposed to 5% Trp after 12 weeks of treatment (Fig. 2b,c). No significant difference from control group was noted in incorporation of 3H-thymidine as shown by autoradiography in gastrocnemius muscle (Fig. 2d,e). Immunohistochemistry showed numerous TNF- α positive cells, indicative of macrophage, in frozen sections of muscle of animals treated with Trp or with p-CPA (Fig. 2f). The levels of TNF- α secreted by cultured spleen macrophages from treated animals were also shown to increase compared to control (Fig. 3). Table 2 summarizes morphological changes in the lung. Focal area of inflammation, increased number of mast cells, eosinophils and TNF- α positive cells accompanied by increased fibrosis, obstruction of alveoli and blood congestion were observed in the lungs (Fig. 4a,b,e,d). A comparison between the groups showed that the frequency of histological changes was enhanced with increased treatment duration or cumulative dose of Trp. However, the same changes were observed after injection of p-CPA alone, but such changes were milder. Autoradiogram of lung tissue

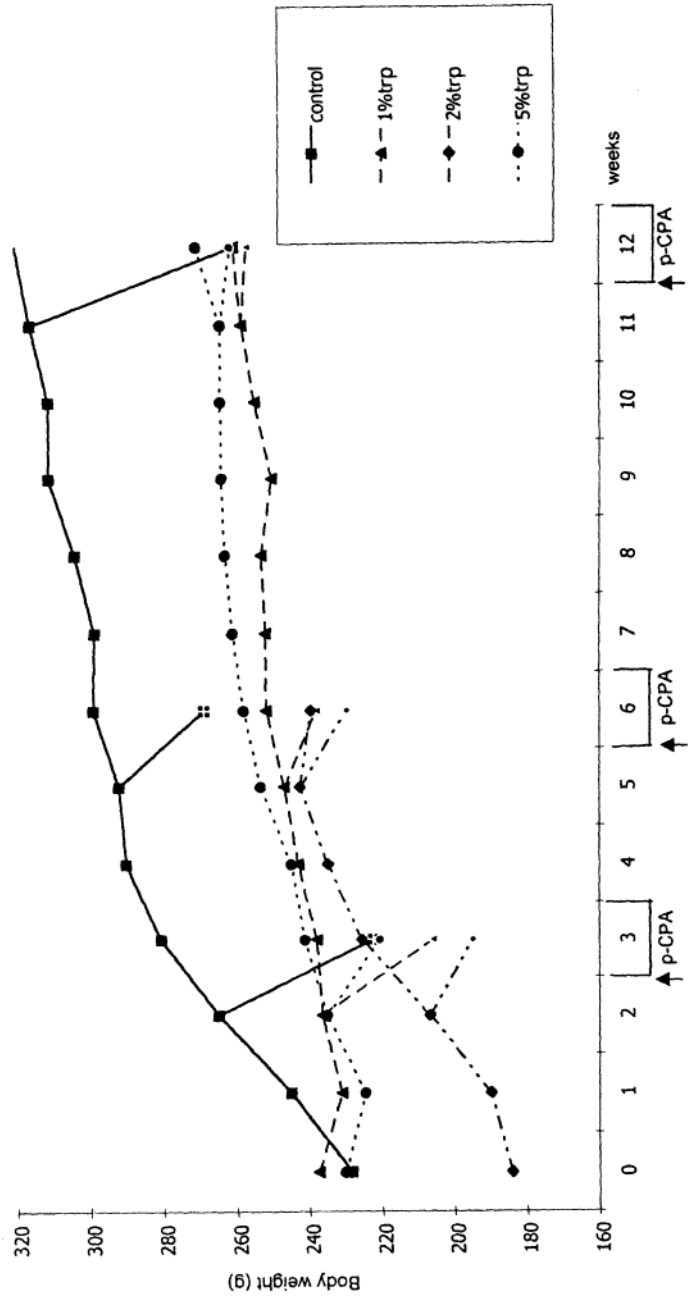


Figure 1. Body weight of rat fed control and L-Trp diets for 3, 6, 12 weeks. Note: decrease body weight after injection of CPA during the last week of treatment.

Table 1. Summary of pathological changes in gastrocnemius.

Duration of treatment				
DIET	2 days	3 weeks	6 weeks	12 weeks
Control	+	+	+	+
	*	*	*	*
		#		
p-CPA		+++	+++	++
		**	*	**
		\$	\$	
		##		
1% Trp	+	+	++	+
	**	**	*	*
		\$	\$	
		###		
1% Trp + p-CPA		+++	++	+++
		***	**	*
		##		
2% Trp	+	++	++	
	**	*	**	
		#		
2% Trp + p-CPA		++	+++	
		*	**	
		#		
5% Trp		++		+++
		**		*
5% Trp + p-CPA		+++		+++
		***		*
	Rare	Moderate	Frequent	
Fibroblasts	+	++	+++	
Mast cells	*	**	***	
Eosinophils	\$	\$	\$	
Macrophages	#	##	###	

showed marked increase of 3H-thymidine labeled cells in the Trp treated group or after p-CPA injection (Fig. 4c,d).

4. DISCUSSION

Many surveys have confirmed the association of “natural” with healthy and safe. The present study demonstrated that a natural compound, L-Trp, when used in high dose, might induce pathologic changes. Previous study demonstrated that increased plasma levels of serotonin (5HT) resulted from dietary-rich tryptophan supplementation (Ronen *et al.*, 1996). Since lung is an active site for removal of 5HT from the plasma, the endothelial cells and lung alveoli are exposed to high concentration of 5HT. Thus, it is conceivable that the high level of 5HT in this tissue could induce the pathological changes that were observed in our study. Furthermore, endothelial thickening and fibrosis were reported in cases of carcinoid syndrome (Freis *et al.*, 1973) and in cases of EMS (Smith *et al.*, 1990). Pakala (1994) reported that 5HT have mitogenic and proliferative effect on endothelial cells by activation different peripheral 5HT receptors. Increased 5HT con-

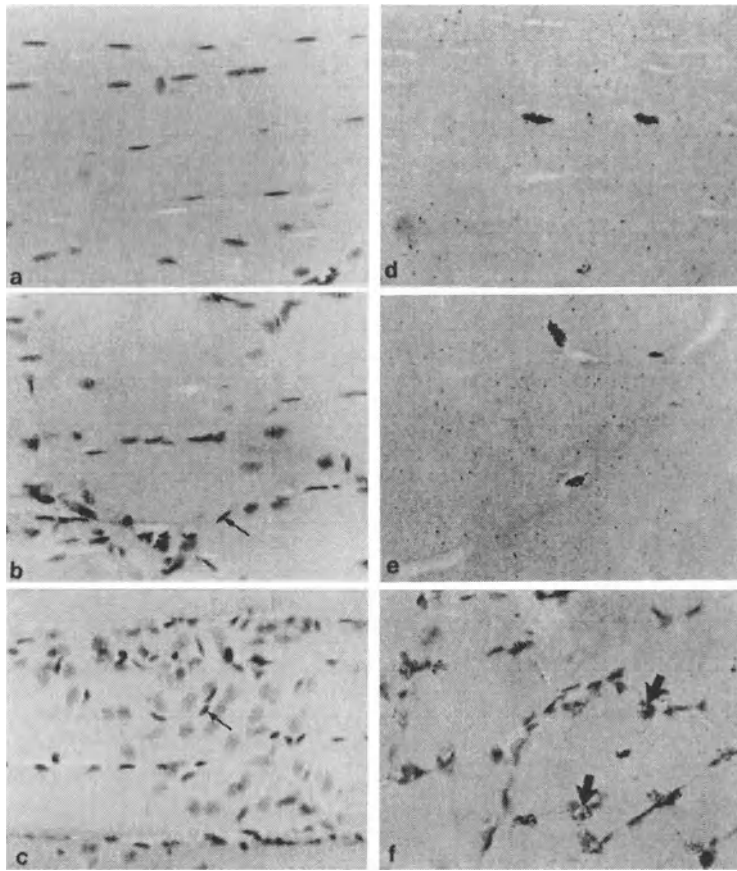


Figure 2. The morphological appearance of gastrocnemius muscle from (a) control, (b) Following 1% Trp for 3 weeks. (c) Following 5% Trp for 12 weeks. Note: inflammatory cells infiltration and an increased number of fibroblasts, small arrow in b and c. H&E $\times 142$. (d) and (e) An autoradiogram after ^3H -thymidine incorporation obtained from (d) control and (e) 1% Trp + p-CPA. (f) Immunohistochemistry detected TNF- α localization following 1% Trp + p-CPA. Note numerous positively labeled cells (arrows) $\times 360$.

centration reflects a continuing process of platelet aggregation and release of cytokines. Recent studies suggested that 5HT had a growth stimulating effect on fibroblasts and vascular smooth muscle cells in lung (Lee *et al.*, 1994). Young (1995) reported that 5HT regulate immune activities through several different 5HT receptors. It enhanced T-cell blastogenesis and B-cell proliferation through 5-HT₁ receptor and inhibit synthesis of TNF- α by macrophages through the 5-HT₂ receptor. 5HT can regulate T cells in a biphasic manner, with low levels being stimulatory to T cells and high levels being suppressive. The present study showed that treatment with excessive L-Trp or with p-CPA alone caused an increase of immunohistochemical positive TNF- α reaction in muscle and lung. Inhibition of tryptophan hydroxylase by p-CPA enhanced Trp levels in plasma and induced its IDO. IDO is an inducible enzyme with high affinity for Trp that is found in blood monocytes, macrophages, fibroblasts, and certain other cell types. Degradation of Trp to indole derivatives has a potent chemotactic activity for blood cells. It is unknown

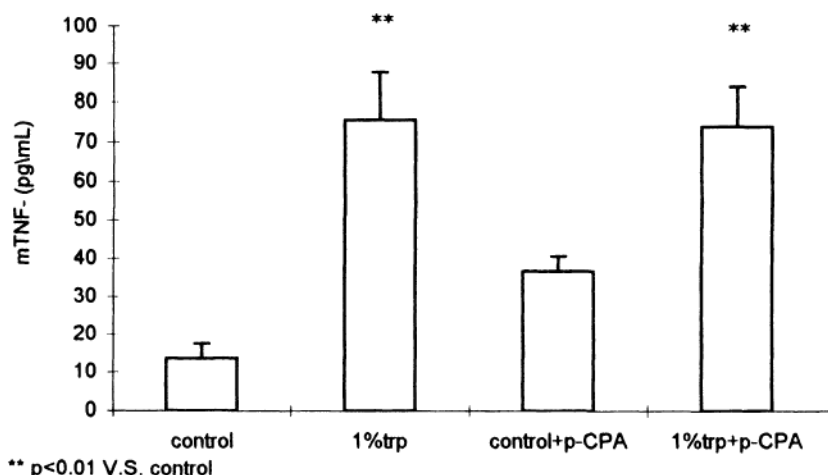


Figure 3. TNF- α secretion content from spleen macrophages from rats fed 1% Trp with and without p-CPA.

if activation of IDO is an epiphenomenon associated with lung inflammatory or fibrotic process or it is important in the pathogenesis of interstitial lung disease (Meyer *et al.*, 1995). Previously it was shown that treatment with p-CPA induced low levels of 5HT in plasma and activation of the kynurenine pathway (Ronen *et al.*, 1996). It seems that some of the metabolites along this pathway are immune-modulators (Saito *et al.*, 1993; Heyes *et al.*, 1992). Picolinic acid, which is produced in this pathway, is a co-factor in macrophage activation (Varesio *et al.*, 1990). The mechanism by which picolinic acid activates macrophages is speculative at the present time. Induction of rRNA in the macrophage may be involved in activation of macrophages to a cytotoxic stage. Amplification of the kynurenine pathway induced inflammation and production of the cytokine TNF- α in muscle and in lung tissue. Positive TNF- α cells are indicative of macrophages. Our results support some of others experimental studies that showed mild but significant morphological changes in Lewis rats fed a diet containing contaminant Trp and pure Trp (Love *et al.*, 1993). It was shown that excessive L-Trp at long term administration with p-CPA injection lower body weight, induce severe pathological changes in lung, spleen and muscle and may even cause death. Anderson and Cowen (1991) have reported a relationship between eating disorders, weight control and Trp. We found that although food consumption in control and in all the experimental groups was the same, the growth patterns in these groups appeared to be different. The weight gain was less in L-Trp treated group and a loss of weight was observed after injection of p-CPA. It is known that Trp can stimulate central serotonin synthesis and induce a rise of growth hormone and prolactin. It seems that inhibition of serotonin synthesis by injection of p-CPA and activation of TDO enzyme, which degrade the essential amino acid Trp, induced relatively malnourished state that possibly can induce weight loss. The interesting observation was that combined treatment of excessive Trp and p-CPA induced high mortality. It was reported previously that quinolinic acid, a metabolite of Trp via kynurenine pathway, is an excitotoxic N-methyl-D-aspartate (NMDA) receptor agonist that is known to play a role in apoptotic process (Heyes *et al.*, 1992; Leonhardt *et al.*, 1986). It is speculated that high levels of Trp with injection of p-CPA probably enhance concentration of quinolinic acid and promote excitotoxicity. We also observed previously that the combined treat-

Table 2. Summary of pathological changes in lung.

Duration of treatment				
DIET	2 days	3 weeks	6 weeks	12 weeks
Control	+	+	+	+
	#	#	*	
			##	
			%	
p-CPA		++	++	+++
		***	**	**
		\$	\$	%
		###	##	
		%%	%	
1% Trp	+	+++	+++	+++
	*	\$	*	***
	#	##	\$	%%%
	%%		###	
			%	
1% Trp + p-CPA		+++	+++	+++
		***	*	*
		\$	\$	%%%
		###	##	
		%%	%%%	
2% Trp	+	++	++	
	**	***	*	
	##	###	\$	
	%%	%%%	###	
2% Trp + p-CPA		+++	+++	
		#	*	
		%%%	\$	
			###	
			%	
5% Trp		+++		+++
		***		*
		%%		%%%
5% Trp + p-CPA		+++		+++
		***		*
		%%		%%%
	Rare	Moderate	Frequent	
Fibroblasts	+	++	+++	
Mast cells	*	**	***	
Eosinophils	\$	\$	\$	
Blood congestion	%	%%	%%%	
Macrophages	#	##	###	

ment of Trp and p-CPA might induce oxidative damage in muscle by release of free radicals. (Livne *et al.*, 1998).

It is concluded that cumulative dose of L-Trp and the duration of exposure appears to be important in induction of pathological changes in some tissues. Induction of kynurenine pathway by injection of p-CPA augments some of the pathological changes and might increase mortality rate. The present observation also confirms previous literature postulating a role of Trp and its metabolites in induction of fibrosis and inflam-

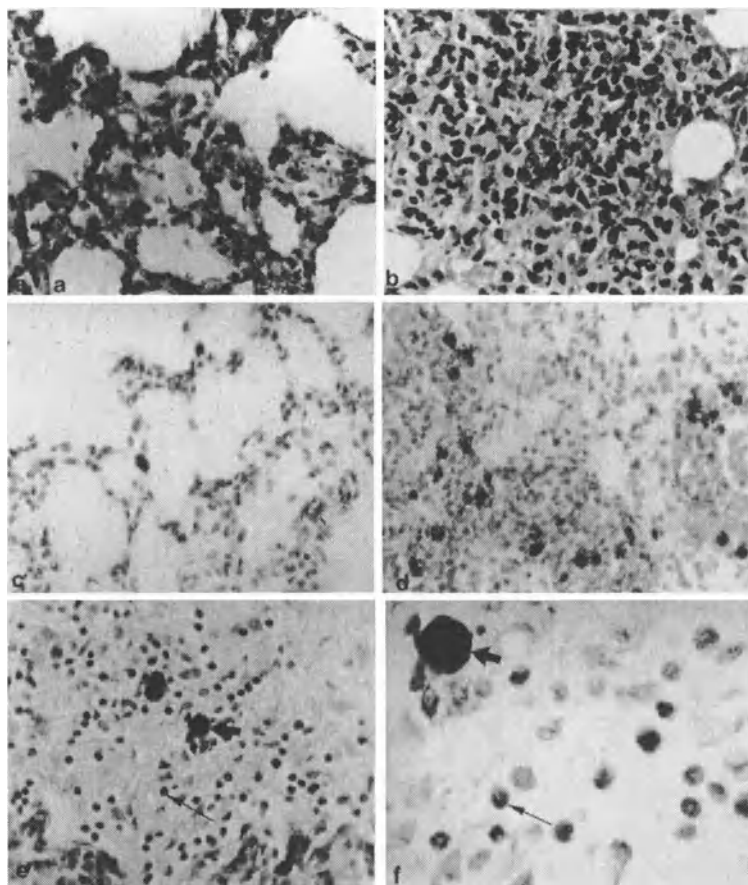


Figure 4. Morphological changes in lung from control (a), treated with 5% Trp for 12 weeks (b). Note: excessive obstruction of alveoli and interstitial fibrosis. H&E $\times 360$. (c) An autoradiogram in control rat, and after 1% Trp + p-CPA treated animal (d). Note: excessive positively 3H-tymidine incorporation and mitosis. $\times 360$. (e) Morphological appearance of a lung treated with 5% Trp and stained with Gimsa. Mast cells (large arrow) and numerous eosinophils (small arrow) are noted in (e) and in (f), a large magnification. $\times 720$.

matory reaction. Further investigation is needed to establish which of the kynurenine pathway metabolites play a significant role in induction toxicity.

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